

ARM μ C KIT WITH USB

- * NXP LPC 2148 Based
- * USB 2.0 with Virtual COM port Drivers
- * 16 x 2 backlit LCD
- * 24 Keys Keyboard
- * 7 Segment LED display
- * Sample 'C' Codes

UNIVERSAL IC TESTER

With In-Circuit IC Testing



One System Many Functions

- ▲ IC Tester to test ICs upto 40 pins (Both Analog and Digital)
- ▲ μ C Development System
- ▲ Word Processor
- ▲ Simple Cable Tester

MME - UIT 40 functionally tests a wide variety of Analog / Linear ICs including ADC, DAC, DPM ICs, Op Amps, Transistor arrays, waveform generators and Digital ICs including μ Ps, μ Cs, EPROMs, RAMs, Peripheral devices apart from 74 series of TTL ICs and 40/45 series of CMOS ICs.

OTHER IC TESTER MODELS

- MME-ADIT 40 Analog & Digital IC Tester
- MME-DIT 2040 Digital IC Tester
- MME-LICT 20 Linear IC Tester
- MME-DIT 20 20 pin IC Tester
- MME-ECT 20 Electronic Components Tester
- MME-HDIT 20 Handheld Digital IC Tester

OTHER PRODUCTS

- * Microprocessor Kits 8085, 8086, Z80
- * Microcontroller Kits 8051, 68HC11, 8096
- * Wide range of educational Interface Cards for ADC, DAC, Stepper Motor, DC Motor, Elevator interface, Logic Controller, Traffic lights, Relay Card, 7 Segment LED display, Alphanumeric LCD, Keyboard etc.,

Advanced ... Yet easy to use Systems from

MME

MINMAX ELECTRONICS

87/1, 1-A Cross, 5th Main,

Domlur Second Stage, Bangalore-560 071

Phone : 2535 3085 Fax : 080 - 2535 2391

E-mail : minmax@vsnl.com

www.minmax.co.in

How to identify the collector and emitter of a transistor?

Dev Kalyan
Kurukshehra

A2. Lead identification of transistors is based on their technical outline (TO). Arrows are not there for

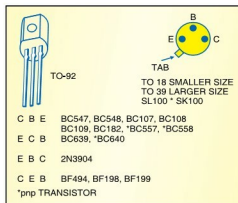


Fig. 3: Identification of leads by TO

identification; perhaps you are referring to the tab on the rim of the transistor available in metal encapsulation like SL100 and SK100.

The lead near the tab, as shown in the diagram, is emitter. Base is in the middle and opposite to emitter is collector.

There is no universal format for the identification of leads. One has to refer to the data sheet. Lead arrangements of some popular transistors available in TO-92

and TO-18 packages are given in Fig. 3.

Q3. I want to know about pin configuration of a seven-segment display having 10 pins.

Jana Sudhar
Nellore

A3. One common requirement for different digital devices is a visual numeric display. Individual LEDs can of course display the binary states of a set of latches or flip-flops. However, for display of decimal numbers, a device is needed,

and seven-segment display meets this requirement. In seven-segment LED, common technique is to use a shaped piece of translucent plastic to operate as a specialised optical fibre, to distribute the light from the LED evenly over a fixed bar shape. The seven bars

are laid out as a squared-off figure '8'. The displays are available in two configurations:

Common Anode. Anodes of all the segments are tied together to the positive bus while respective segment cathode is grounded for switching the segment on.

Common Cathode. Cathodes of all the segments are tied together and grounded while respective segment anode is connected to the positive bus for switching the segment on.

Segment bars are named as a, b, c, d, e, f, g and dp. The pin layout of both the configurations are given in Fig. 4.

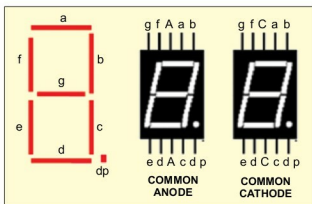


Fig. 4: Pin layout

Testing of display. The serviceability/configuration can be checked by conducting normal diode check between centre pin A or C and different bars. Commonly available displays are:

LTS542 common anode

LTS543 common cathode

Answers compiled by Col (Retd) N.C. Pande

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender.